| Basic Differentiation Rules | Basic Integration Rules |
|---|---|
| $\frac{d}{dx}C = 0$ | $\int 0\,dx = C$ |
| $\frac{d}{dx}(kx) = k$ | $\int k\,dx = kx + C$ |
| $\frac{d}{dx}[kf(x)] = kf'(x)$ | $\int [kf(x)]\,dx = k\int f(x)\,dx$ |
| $\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$ | $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$ |
| $\frac{d}{dx}x^n = nx^{n-1}$ | $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C,\ n \neq -1$ (Power Rule) |
| $\frac{d}{dx}\sin x = \cos x$ | $\int \cos x\,dx = \sin x + C$ |
| $\frac{d}{dx}\cos x = -\sin x$ | $\int \sin x\,dx = -\cos x + C$ |
| $\frac{d}{dx}\tan x = \sec^2 x$ | $\int \sec^2 x\,dx = \tan x + C$ |
| $\frac{d}{dx}\cot x = -\csc^2 x$ | $\int \csc^2 x\,dx = -\cot x + C$ |
| $\frac{d}{dx}\sec x = \sec x \tan x$ | $\int \sec x \tan x\,dx = \sec x + C$ |
| $\frac{d}{dx}\csc x = -\csc x \cot x$ | $\int \csc x \cot x\,dx = -\csc x + C$ |
| $\frac{d}{dx}e^x = e^x$ | $\int e^x\,dx = e^x + C$ |
| $\frac{d}{dx}\ln x = \frac{1}{x},\ x > 0$ | $\int \frac{1}{x}\,dx = \ln\lvert x\rvert + C$ |

Other tips:

- A function is concave up on the interval $(a, b)$ if $f''(x) > 0$ for all $x$ in $(a, b)$.
- A function is concave down on the interval $(a, b)$ if $f''(x) < 0$ for all $x$ in $(a, b)$.
- If $f(x)$ is a continuous function on the interval $[a, b]$ and $F'(x) = f(x)$, then $\int_a^b f(x)dx = F(b) - F(a)$.
- If $\lim_{x \to a^\pm} f(x) = \pm\infty$, then $x = a$ is a vertical asymptote.
- If $\lim_{x \to \pm\infty} f(x) = c$, where $c$ is a finite number, then $y = c$ is a horizontal asymptote.
- The point-slope equation of a line with slope $m$ and through the point $(x_1, y_1)$ is $y = y_1 + m(x - x_1)$.
- $\frac{1}{x^n} = x^{n-1}$ and $x^{1/m} = \sqrt[m]{x}$.
- You might need the product rule: $\frac{d}{dx}f(x)g(x) = f'(x)g(x) + g'(x)f(x)$

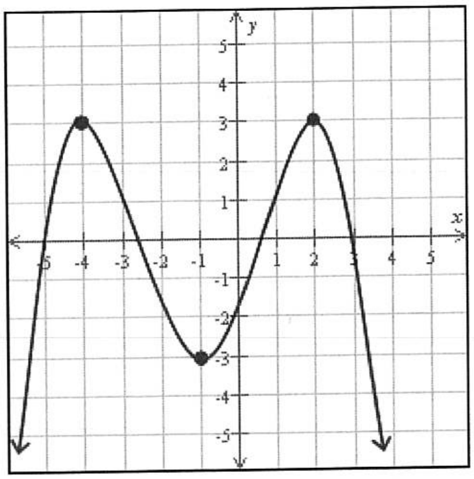
# Retention Exam III #1

Student Nam[redacted]
Number of Questions: 9

Class Name: **Calculus 1 - fall 2025**
Instructor Name [redacted]

## Question 1 of 9

Here is a graph of the function $g$.




Use the graph to find the following.
If there is more than one answer, separate them with commas.

All local maximum values of $g$: 3

All values at which $g$ has a local maximum: (-4,3), (2,3)

## Question 2 of 9

Consider the function defined by the equation below.

$$y = 6 - 2x^2 - 2x$$

Find the absolute minimum and absolute maximum values of $y$ on the interval $[-4, 1]$. Give exact answers, not decimal approximations. If there is no absolute minimum or absolute maximum value, write "None".

| | |
|---|---|
| Absolute minimum value: | -18 |
| Absolute maximum value: | $\frac{13}{2}$ or 6.5 |

$6-2x^2-2x$

$-4x-2=0$

$\frac{-4x}{-4} = \frac{2}{-4}$

$x = -\frac{1}{2}$ CRIT POINT

$6-(2(-4)^2)$

$2(16)$

$6-32-2(-4)$

$6-32+8$

$\boxed{-18}$

$6-2(1)^2-2(1)$

$6-2-2$

$\boxed{y=2}$

$6-2(-\frac{1}{2})^2$

$-2(\frac{1}{4})$

$6-\frac{1}{2}+1$

$6\frac{1}{2}$

## Question 3 of 9

Consider the function $f$ given below.

$$f(x) = -1 - x + x^2 - 3x^2$$

$f'(a)(x-a) + f(a)$

Find $L(x)$, the linearization of $f$ at the value $a = 0$.

$f(0) = -1 - 0 + 0^2 - 3(0)^2$

$L(x) = -1(x-0) - 1$

$-1 - 0 + 0 - 0$

$f'(a) = -1 + 2x - 6x$

$f(0) = -1$

$f'(0) = -1 + 0 - 0$

$f'(0) = -1$

## Question 4 of 9

Find the following limit. If necessary, select the most informative answer from $-\infty$, $\infty$, and "Does Not Exist."

$$\lim_{t \to -\infty} \frac{t^2 - 1}{t^3 - 7} = -\infty$$

$\frac{-\infty^2 - 1}{-\infty^3 - 7} \quad \frac{\infty}{-\infty} = -\infty$

## Question 5 of 9

Let the function $f$ be defined as follows.

$$f(x) = 2x^3 + 9x^2 + 12x - 5$$

Find the critical numbers (critical points) of the function. Use exact values (not decimal approximations). If there is more than one number, separate them with commas. Click on "None" if applicable.

$6x^2 + 18x + 12 = f'(x)$

Critical numbers of $f$: $-2, -1$

## Question 6 of 9

Let the function $f$ be defined as follows.

$$f(x) = x^3 + 9x^2 + 8x - 5$$

(a) Write your answers as an interval or list of intervals. When writing a list of intervals, make sure to separate them with commas and to use as few intervals as possible. Write "None" if applicable.

Write each answer as an interval or a list of intervals. When writing a list of intervals, make sure to separate them with commas and to use as few intervals as possible. Click on "None" if applicable.

Interval(s) on which $f$ is concave up: $(-3, \infty)$

Interval(s) on which $f$ is concave down: $(-\infty, -3)$

(b) Find all the inflection points (points of inflection) of the function $f$. Write the points as ordered pairs. Use exact values in your answers (not decimal approximations).

If there is more than one point, separate them with commas. Click on "None" if applicable.

All inflection points of $f$: $(-3, 25)$

## Question 7 of 9

Find the integral.

$$\int \left(-4\sqrt[3]{t} + \frac{1}{2\sqrt[3]{t}}\right) dt = -3t^{4/3} + \ln|2\sqrt[3]{t}| + C$$

$-4t^{1/3} + \frac{1}{2t^{1/3}}$

$\frac{-4t^{4/3}}{4/3}$ $\quad -4\,\frac{3}{4}$ $\quad \frac{-12}{4} =$ $\quad -3t^{4/3}$ $\quad \ln$

## Question 8 of 9

Evaluate the integral. Give an exact answer and not a decimal approximation.

$$\int_{-2}^{3}\left(4x^2 - 3x - 5\right)dx = \underline{\quad \frac{85}{6} \quad}$$

$\frac{4}{3}x^3 - \frac{3}{2}x^2 - 5x \Big|_{-2}^{3}$

$-\frac{32}{3} + \frac{12}{3} = -\frac{20}{3}$

$\frac{4}{3}(3)^3$ $\quad$ $\frac{4}{3}(-8) - \frac{3}{2}(4) + 10$

$\frac{4}{3}(27) - \frac{3}{2}(9) - 15$ $\quad$ $-\frac{32}{3} - 6 + 10$

$36 - \frac{27}{2} - 15$ $\quad$ $\frac{42}{2} - \frac{27}{2} = \frac{15}{2} - \left(-\frac{20}{3}\right)$

$\frac{45}{6} + \frac{40}{6}$

## Question 9 of 9

Let the function $f$ be defined as follows.

$$f(x) = -2x^3 + 18x^2 - 30x + 7$$

(a) Determine the interval(s) in the domain of $f$ on which the function is increasing and the interval(s) in the domain of $f$ on which the function is decreasing.

Write each answer as an interval or a list of intervals. When writing a list of intervals, make sure to separate them with commas and to use as few intervals as possible. Write "None" if applicable.

Interval(s) on which $f$ is increasing: $(1, 5)$

Interval(s) on which $f$ is decreasing: $(-\infty, 1), (5, \infty)$

(b) Find all the points at which the function $f$ attains a local maximum and all the points at which $f$ attains a local minimum.

Write the points as ordered pairs. If there is more than one point, separate them with commas. Write "None" if applicable.

All local maximum points of $f$: $(5, 57)$

All local minimum points of $f$: $(1, -7)$

LUCAN RHATIGAN

$6x^2+18x+12=f'(x)$

$6x^2+18x+12=0$

$6(x^2+3x+2)=0$

$(x+2)(x+1)$

$x=-2 \quad x=-1$

39

27+54+8

$f'(x)\ 3x^2+18x+8=0$

$3(x^2+6x$

$\frac{4}{3}+\frac{2}{3} \quad \frac{8}{3}$

CANT FACTOR

$6+4 \quad \frac{10}{3}$

IP

$6x+18=0$

$\frac{6x}{6}=\frac{-18}{6}$

$\boxed{x=-3}$

IP

$-2x^3+18x^2-30x+7$

$-6x^2+36x-30$

$-6(x^2-6x+5)=0$

$(x-5)(x-1)$

$x=5 \quad x=1$ CRIT POINTS

$3x^2+18x=-8$

5

$x(3x+18)=-8$

$-6(4)^2+36(4)-30=+$

$-2(5)^3+18(5)^2-30(5)+7$

$-6(0)^2+36(0)-30=-$

$-250+450$

$200-150+7$

$=57$

$-2(1)+18(1)-30(1)+7$

$16-30$

$-14+7=-7$

$-3^3+9(-3)^2+8(-3)-5$

$-27+81-24-5$

$=25$

$\boxed{IP = -3, 25}$